

7. (a) State **three** assumptions of the kinetic theory of ideal gases. [3]

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- (b) A container of fixed volume contains oxygen gas. Five molecules in the gas have speeds 480, 521, 436, 445 and 503 m s^{-1} .

- (i) Determine the rms speed of these five molecules. [2]

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- (ii) Determine the temperature of a sample of oxygen gas whose molecules have this rms speed. (Relative molecular mass of oxygen = 32.) [3]

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- (c) Gareth states that when the temperature of the gas is doubled both the rms speed of the molecules and the pressure are doubled. Determine whether or not Gareth is correct.

[4]

Examiner
only

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